

AMAZON WEB SERVICES • CLOUD PRACTICAL

S3 Lifecycle Policy & Cost Optimization

Managing Data Lifecycle with Automated Transitions

Shreya Panchal | Assignment: 13 lifecycle policy and cost optimization managing data lifecycle with automated transitions.
Implement lifecycle policies to move objects from s3 to glacier.

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ASSIGNMENT	Assignment 13
FOCUS AREA	Amazon S3 Lifecycle Rules & Storage Cost Optimization
PLATFORM	AWS Management Console & AWS Pricing Calculator

Submitted as part of Cloud Computing practical coursework

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01	Create an S3 bucket and upload sample restaurant images
02	Configure a lifecycle rule to transition objects to Glacier after 30 days
03	Compare S3 Standard vs Glacier storage costs using the AWS Pricing Calculator
04	Add a lifecycle rule to auto-delete 'offers/' prefixed objects after 60 days
05	Generate and explain a JSON lifecycle policy configuration

TASK 01 / 05

AMAZON S3 LIFECYCLE & COST OPTIMIZATION

QUESTION

1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

ANSWER

Answer: A new Amazon S3 bucket named **my-zomato-images** was created successfully in the **Asia Pacific (Mumbai)** region. Three sample restaurant menu images were uploaded using the AWS Management Console.

Uploaded Files:

- **masala-dosa.jpg** – Image of a South Indian Masala Dosa.
- **paneer-pizza.jpg** – Image of a Paneer Pizza.
- **veg-burger.jpg** – Image of a Vegetarian Burger.

Steps Performed:

1. Logged in to the AWS Management Console.
2. Opened the **Amazon S3** service.
3. Clicked **Create bucket**.
4. Entered the bucket name **my-zomato-images**.
5. Selected the **Asia Pacific (Mumbai)** region.
6. Created the bucket with default settings.
7. Opened the bucket and clicked **Upload**.
8. Selected the three restaurant menu images.
9. Uploaded the files successfully and verified that all images were available in the bucket.

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Asia Pacific (Mumbai) ap-south-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)
my-zomato-images

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (3 total, 814.1 KB) [Remove](#) [Add files](#) [Add folder](#)

All files and folders in this table will be uploaded.

☒	Name	Folder	Type	Size
☒	veg-burger.jpg	-	image/jpeg	426.3 KB
☒	paneer-pizza.jpg	-	image/jpeg	137.4 KB
☒	masala-dosa.jpg	-	image/jpeg	250.3 KB

Destination [Info](#)

Destination

RESULT:

The S3 bucket **my-zomato-images** was successfully created, and the three restaurant menu images were uploaded successfully.

TASK 02 / 05

AMAZON S3 LIFECYCLE & COST OPTIMIZATION

QUESTION

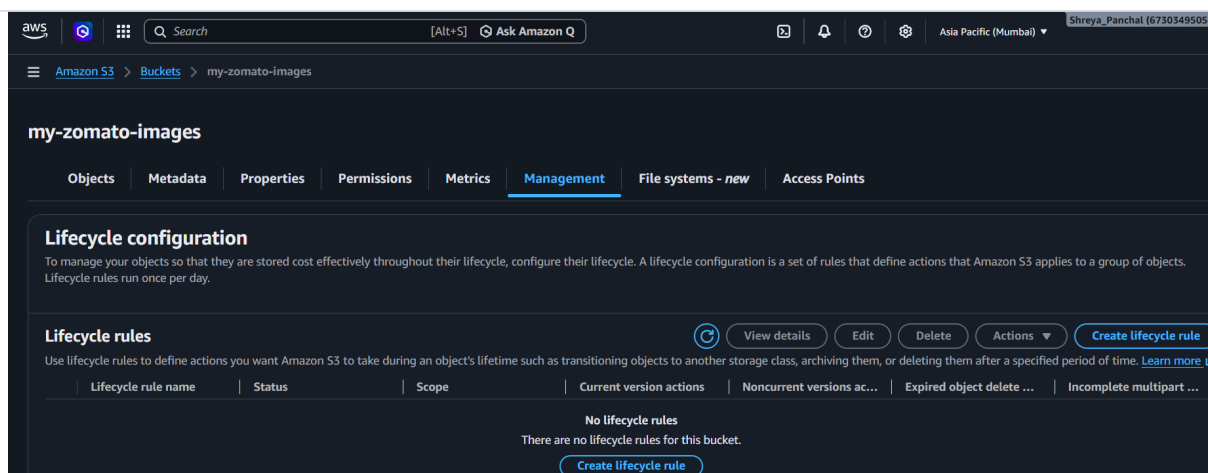
2. Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class. **Hint:** Use the S3 Management Console's 'Lifecycle rules' section to define the transition action.

ANSWER

Answer: A lifecycle policy was created on the **my-zomato-images** S3 bucket to automatically transition all objects to the **Glacier Instant Retrieval** storage class after **30 days**. This helps reduce storage costs while keeping the data securely archived.

Steps Performed:

1. Logged in to the AWS Management Console.
2. Opened the **Amazon S3** service.
3. Selected the **my-zomato-images** bucket.
4. Opened the **Management** tab.
5. Clicked **Create lifecycle rule**.
6. Entered the rule name **Move-To-Glacier-After-30-Days**.
7. Selected **Apply to all objects in the bucket**.
8. Enabled **Transition current versions of objects between storage classes**.
9. Configured the transition to move objects to **Glacier Instant Retrieval** after **30 days**.
10. Saved the lifecycle rule successfully.



aws Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Shreya_Panchal (ex)

Amazon S3 > Buckets > my-zomato-images > Lifecycle configuration > Create lifecycle rule

Create lifecycle rule [Info](#)

Lifecycle rule configuration

Lifecycle rule name

Move-To-Glacier-After-30-Days

Up to 255 characters

Choose a rule scope

☐ Limit the scope of this rule using one or more filters

☒ Apply to all objects in the bucket

⚠ Apply to all objects in the bucket
If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

☒ Transition current versions of objects between storage classes

Amazon S3 > Buckets > my-zomato-images > Lifecycle configuration > Create lifecycle rule

Lifecycle rule actions

Choose the actions you want this rule to perform.

☒ Transition current versions of objects between storage classes
This action will move current versions.

☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.

☐ Expire current versions of objects

☐ Permanently delete noncurrent versions of objects

☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

⚠ Transitions are charged per request
For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the **Storage**

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

ℹ By default, objects less than 128KB will not transition across any storage class
We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, you must use a size filter for each applicable lifecycle rule with a transition action.

Amazon S3 > Buckets > my-zomato-images > Lifecycle configuration > Move-To-Glacier-After-30-Days

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Glacier Instant Retrieval

Days after object creation

30

[Remove](#)

[Add transition](#)

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

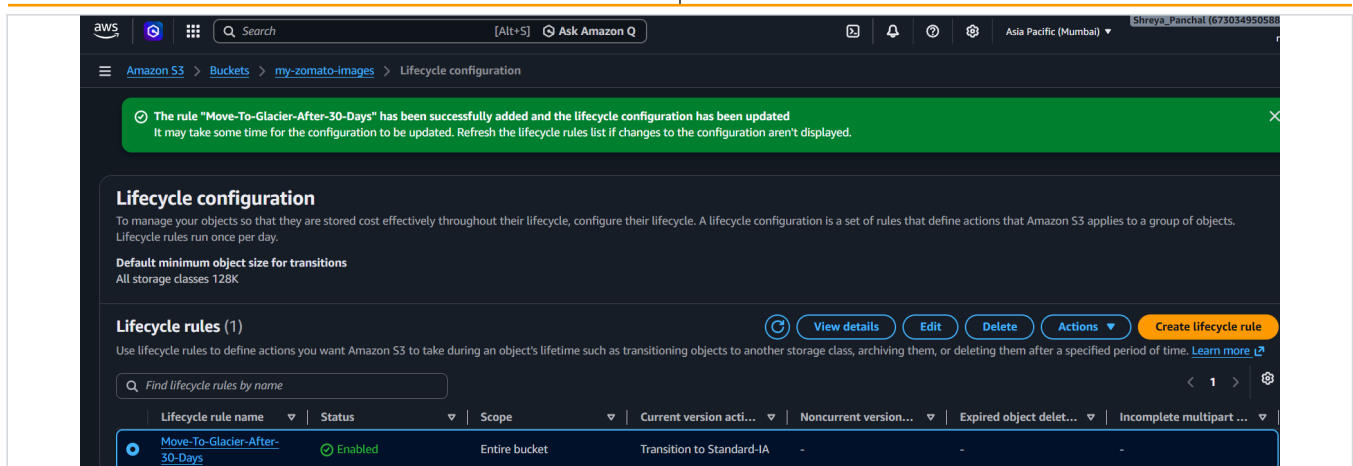
Day 30

- Objects move to Glacier Instant Retrieval

Noncurrent versions actions

Day 0

No actions defined.

**RESULT:**

The lifecycle policy was successfully created and enabled. All objects in the **my-zomato-images** bucket will automatically transition to the **Glacier Instant Retrieval** storage class after **30 days**, helping optimize long-term storage costs.

TASK 03 / 05

AMAZON S3 LIFECYCLE & COST OPTIMIZATION

QUESTION

3. Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator.

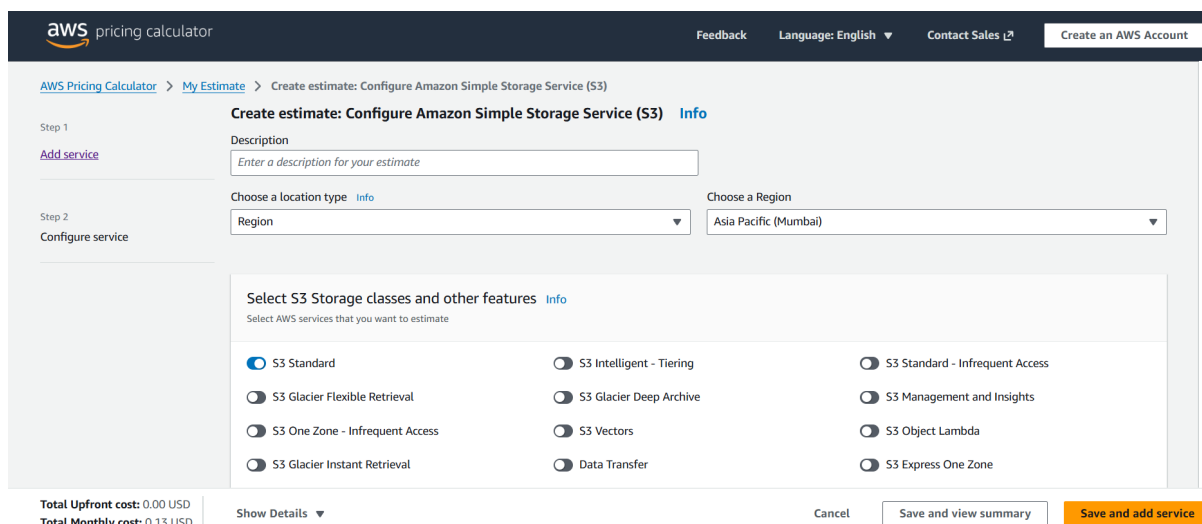
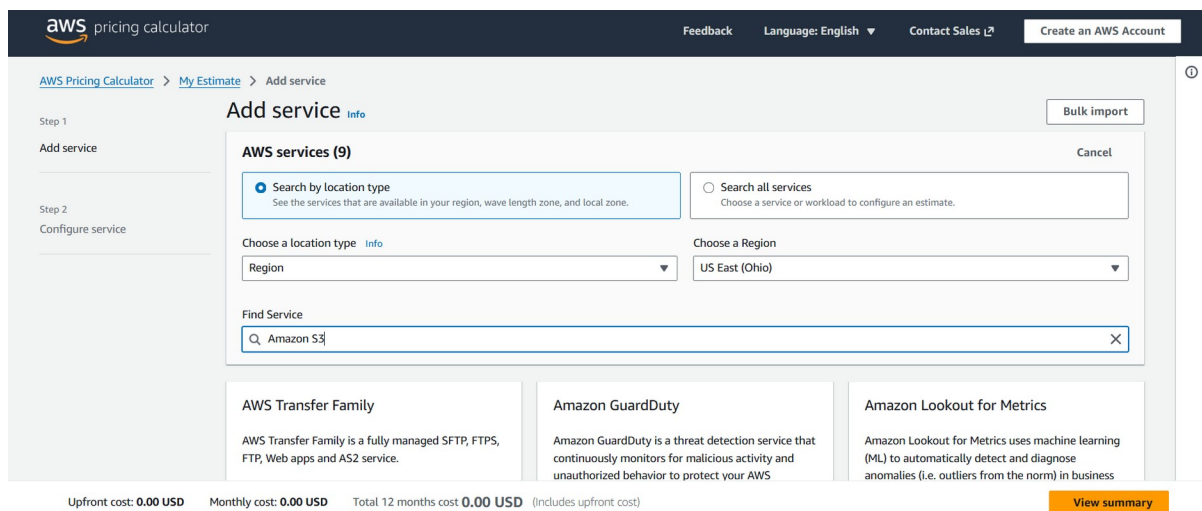
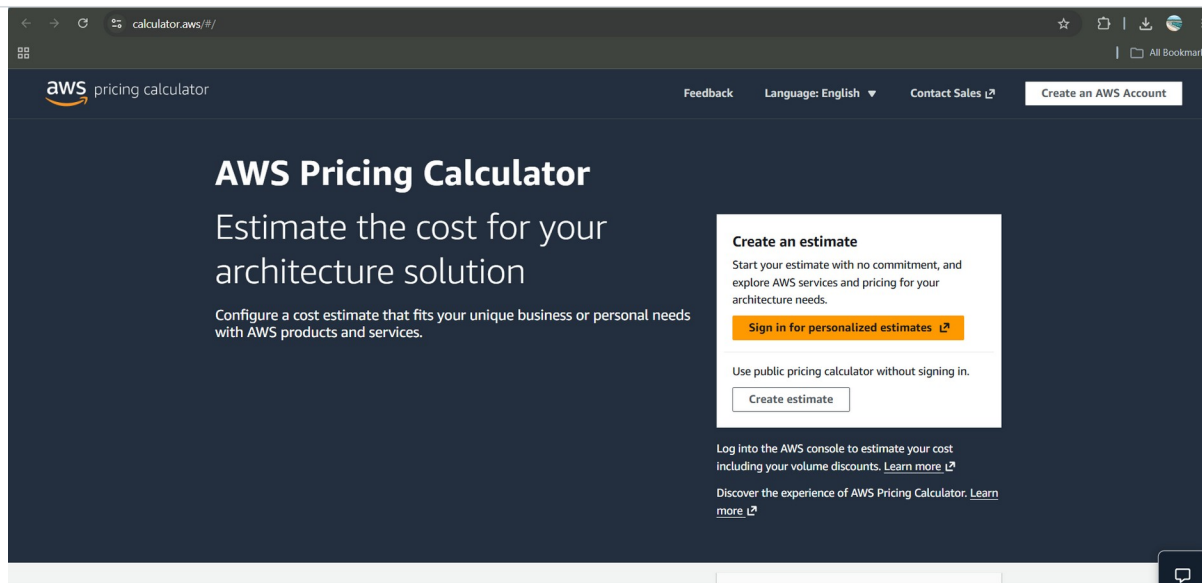
Hint: Use the AWS Pricing Calculator website and note down the difference in INR.

ANSWER

Answer: The AWS Pricing Calculator was used to estimate the monthly storage cost for storing **5 GB** of image files in **Amazon S3 Standard** and **Amazon S3 Glacier Flexible Retrieval**. The comparison showed that **Glacier** is more cost-effective than **S3 Standard** for long-term storage of infrequently accessed data.

Steps Performed:

1. Opened the **AWS Pricing Calculator** website.
2. Clicked **Create Estimate**.
3. Searched for and selected **Amazon S3**.
4. Selected the **Asia Pacific (Mumbai)** region.
5. Chose **S3 Standard** as the storage class.
6. Entered **5 GB per month** as the storage amount.
7. Left all request and data transfer values at their default settings.
8. Clicked **Add to Estimate** to generate the monthly cost for **S3 Standard**.
9. Added another **Amazon S3** service to the estimate.
10. Selected **S3 Glacier Flexible Retrieval** as the storage class.
11. Entered **5 GB per month** as the storage amount.
12. Left all other settings unchanged and clicked **Add to Estimate**.
13. Compared the estimated monthly costs of **S3 Standard** and **S3 Glacier Flexible Retrieval** on the **My Estimate** page.



aws pricing calculator Feedback Language: English Contact Sales Create an AWS Account

▼ S3 Standard Info

The calculations below exclude Free Tier discounts.

S3 Standard storage 5 Unit GB per month

aws pricing calculator Feedback Language: English Contact Sales Create an AWS Account

Successfully added Amazon Simple Storage Service (S3) estimate.

AWS Pricing Calculator > My Estimate > Add service

Step 1 Add service

Step 2 Configure service

Add service info Bulk import

AWS services (148) Cancel

☒ Search by location type See the services that are available in your region, wave length zone, and local zone.

☐ Search all services Choose a service or workload to configure an estimate.

Choose a location type Info Choose a Region

Region Asia Pacific (Mumbai)

aws pricing calculator Feedback Language: English Contact Sales Create an AWS Account

Successfully added Amazon Simple Storage Service (S3) estimate.

AWS Pricing Calculator > My Estimate > Create estimate: Configure Amazon Simple Storage Service (S3) Info

Step 1 Add service

Step 2 Configure service

Create estimate: Configure Amazon Simple Storage Service (S3) Info

Description Enter a description for your estimate

Choose a location type Info Choose a Region

Region Asia Pacific (Mumbai)

Select S3 Storage classes and other features Info

Select AWS services that you want to estimate

☐ S3 Standard ☐ S3 Intelligent - Tiering ☐ S3 Standard - Infrequent Access

☐ S3 Glacier Flexible Retrieval ☐ S3 Glacier Deep Archive ☐ S3 Management and Insights

☐ S3 One Zone - Infrequent Access ☐ S3 Vectors ☐ S3 Object Lambda

Total Upfront cost: 0.00 USD Total Monthly cost: 0.03 USD Show Details

Cancel Save and view summary Save and add service

S3 Glacier Instant Retrieval feature

▼ S3 Glacier Instant Retrieval Info

S3 Glacier Instant Retrieval storage 5 Unit GB per month

RESULT:

The cost comparison showed that **Amazon S3 Glacier Flexible Retrieval** has a lower monthly storage cost than **Amazon S3 Standard** for storing 5 GB of image files. This demonstrates that Glacier is a better option for long-term storage of data that is rarely accessed, helping reduce overall storage costs.

TASK 04 / 05

AMAZON S3 LIFECYCLE & COST OPTIMIZATION

QUESTION

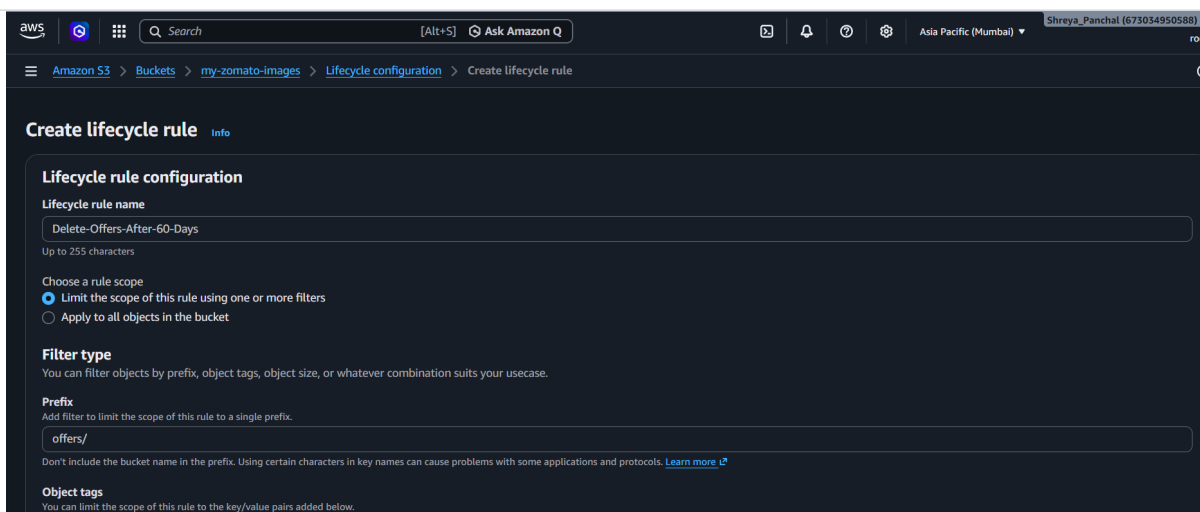
4. Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.

ANSWER

Answer: A new lifecycle rule was created in the **my-zomato-images** S3 bucket to automatically delete all objects stored under the **offers/** prefix after **60 days**.

Steps Performed:

1. Logged in to the AWS Management Console.
2. Opened the **Amazon S3** service.
3. Selected the **my-zomato-images** bucket.
4. Opened the **Management** tab.
5. Clicked **Create lifecycle rule**.
6. Entered the rule name **Delete-Offers-After-60-Days**.
7. Selected **Limit the scope of this rule using one or more filters**.
8. Entered the prefix **offers/**.
9. Selected **Expire current versions of objects**.
10. Configured the expiration period to **60 days after object creation**.
11. Clicked **Create rule** to save the lifecycle policy.



Lifecycle rule actions
Choose the actions you want this rule to perform.

- ☐ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☒ Expire current versions of objects
 - ☐ Permanently delete noncurrent versions of objects
 - ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

Expire current versions of objects
For version-enabled buckets, Amazon S3 adds a delete marker and the current version of an object is retained as a noncurrent version. For non-versioned buckets, Amazon S3 permanently removes the object. [Learn more](#)

Days after object creation
60

Review transition and expiration actions

Current version actions Noncurrent versions actions

Lifecycle configuration
To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (2)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multi...
Move-To-Glacier-After-30-Days	Enabled	Entire bucket	Transition to Glacier Instant f	-	-	-
Delete-Offers-After-60-Days	Enabled	Filtered	Expires	-	-	-

RESULT:

The lifecycle policy was successfully created. Any object uploaded with the prefix **offers/** (for example, **offers/summer2024.jpg**) will be automatically deleted after **60 days**.

TASK 05 / 05

AMAZON S3 LIFECYCLE & COST OPTIMIZATION

QUESTION

5. Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

Answer

Answer: **Sample S3 Lifecycle Policy (JSON)**

```
{
  "Rules": [
    {
      "ID": "MoveToGlacierAndDelete",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 180
      }
    }
  ]
}
```

Explanation of Each Section

JSON Section	Explanation
Rules	Contains one or more lifecycle rules that Amazon S3 will apply.
ID	A unique name used to identify the lifecycle rule.
Status	Enables or disables the lifecycle rule. Here it is set to Enabled .

JSON Section	Explanation
Filter	Specifies which objects the rule applies to. An empty prefix ("") means the rule applies to all objects in the bucket.
Transitions	Defines when objects should move to another storage class.
Days: 15	Objects are automatically moved 15 days after upload .
StorageClass: GLACIER	Moves objects to the Amazon S3 Glacier storage class for lower-cost archival storage.
Expiration	Defines when objects are permanently deleted.
Days: 180	Objects are automatically deleted 180 days after upload .

```

{
  "Rules": [
    {
      "ID": "MoveToGlacierAndDelete",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 180
      }
    }
  ]
}

```

Ln 20, Col 2 | 305 characters | Plain text | 100% | Windows (CRLF) | UTF-8

RESULT:

This lifecycle policy automatically moves all objects to **Amazon S3 Glacier** after **15 days** and permanently deletes them after **180 days**. This helps reduce storage costs while automatically removing old files that are no longer needed.